Existing Problem

- Wind velocity estimation using the 205 MHz ST radar at CUSAT requires accurate identification of peaks in Doppler spectra.
- **Major challenges:**
  - Low signal-to-noise ratio (SNR)
  - Ground clutter and unwanted interference
  - Multiple spectral peaks and ambiguous peak selection
- Incorrect peak detection leads to errors in Doppler velocity estimation and discontinuities in wind profiles.
- **Objective:** Compare MCCF, IME, and hybrid methods to study their accuracy under different radar signal conditions.